

Course Project - Final Submission

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Research Topic

This paper examines the key factors that drive Airbnb performance in Boston using listing-level data. Specifically, it analyzes how pricing strategies and minimum-night requirements influence estimated occupancy and revenue. In addition, the study explores whether the size of a listing, measured through building type and number of bedrooms or beds, affects its revenue-generating ability. Together, these insights aim to help hosts optimize listing decisions to improve financial outcomes. This topic is interesting to me because I have wondered about the viability of using Airbnb to lease houses and longer-term rentals.

Research Questions

How do pricing strategies and minimum night requirements affect occupancy and revenue performance?

Does the neighborhood of an Airbnb listing affect its ability to bring in revenue?

Relevant Theory - Signaling Theory

Signaling Theory provides a useful framework for understanding how Airbnb hosts communicate the quality and value of their listings to potential guests in an environment characterized by information asymmetry. In the short-term rental market, guests cannot directly observe the true quality, comfort, or reliability of a listing before booking. Instead, they must rely on observable details that hosts provide through listing attributes, pricing decisions, and property features. These signals help reduce uncertainty and shape guest expectations, which ultimately influence booking behavior, occupancy rates, and revenue performance. In the context of my research, Signaling Theory helps explain why pricing strategies and minimum-night requirements can meaningfully affect listing performance. A host's chosen nightly rate acts as a signal of quality: higher prices can indicate superior amenities, better maintenance, or a premium location, while lower prices may signal affordability or lower perceived quality. Similarly, minimum-night requirements can operate as a signal of exclusivity or professionalism. Listings that require longer stays may be perceived as more stable, better maintained, or targeted toward a more committed traveler segment. These signals can influence how guests interpret the

value of the listing and whether they decide to book, thus shaping occupancy and revenue outcomes. Overall, Signaling Theory provides a coherent explanation for why observable listing characteristics should strongly correlate with occupancy and revenue. The theory supports the premise that hosts strategically or unintentionally send signals about quality and value through pricing decisions, minimum-stay rules, and physical listing attributes. These signals shape guest perceptions and booking behaviors, thereby providing a theoretical basis for the relationships explored in this study.

Hypotheses

Airbnb listings with high prices and minimum night requirements see higher revenue but are vacant for a relatively longer period than other listings.

(Relationship: Higher prices/min. night requirements have a positive impact on revenue, but negative impact on occupancy.)

Airbnb listings in neighborhoods closer to the center of Boston generate more revenue because of the proximity to tourist activities.

(Relationship: Neighborhood has a statistically significant effect on revenue.)

(Moderating Effect: This effect is weaker in neighborhoods which are farther from the center of Boston.)

Preprocessing

Excel: Removed unnecessary columns

Python: Dropped missing values and changed data type of relevant columns

Key Findings from Data Visualization and Exploration

1. Listings accommodating 6-9 guests and priced between \$600-\$1,000 per night earned the highest average revenue over the last year.
This indicates that optimal revenue occurs at the intersection of medium-large listing size and mid-high pricing, informing future modeling on interaction effects between size and price.
2. Listings with minimum-night requirements under 40 days consistently outperformed those with stricter requirements, with no listing above 40 days

earning more than \$90k.

This shows that stricter stay requirements reduce booking flexibility, which will be important when modeling demand constraints.

3. Downtown Boston recorded both the highest average revenue per listing and the highest total revenue among all neighborhoods.

Location therefore plays a major role in revenue generation, suggesting that neighborhood should be modeled as a key demand predictor.

4. Listings with minimum-night requirements under 10 days had the highest median revenue.

This highlights how flexible booking policies significantly enhance earnings potential and should be included in future regression modeling.

5. Large clusters of listings had 60- or 90-day minimum-night requirements, yet only half of listings above 90 days earned any revenue.

This reveals a supply-demand imbalance in long-term rentals, guiding decisions on whether to treat these listings as a separate market segment in analysis.

6. Single-guest listings generated revenue only 79% of the time, while all other accommodation categories exceeded 83%.

This suggests that one-person listings face lower demand and higher competition, informing occupancy modeling and segmentation.

7. Very large listings (accommodating 12, 14, or 16 guests) were rare but earned revenue at very high rates (92%-100%).

This indicates a profitable niche with low competition, and suggests nonlinear effects of listing size that should be captured in modeling.

8. Smaller listings face more competition but fill more consistently, while larger listings carry higher vacancy risk despite higher revenue potential.

This tradeoff informs strategies around property selection and will guide multi-metric modeling (e.g., revenue + occupancy).

9. Listings accommodating 6-9 guests were occupied only 35-37% of the year on average, despite producing high annual revenue.

This suggests that high nightly rates compensate for lower occupancy, informing separate models for revenue and occupancy rate.

10. Downtown Boston stands out as the strongest location for Airbnb performance, and transit-accessible areas may benefit from spillover demand.

This reinforces location as a dominant driver of revenue and suggests including proximity or neighborhood fixed effects in predictive modeling.

Variable Summary

Variable Type	Variable Categories	Variable	Action	Mean	Std.	Max.	Min.
Dependent	Performance	Revenue L365D	Revenue in the last year, used in raw and in logarithmic form for modeling	29,621.68	36,835.67	344,250.00	0
		Occupancy L365D	Number of booked days in the last year, used to create occupancy rate (divide by 365)	122.96	95.13	255.0	0
Independent	Listing Attributes	Price	Price of listing. Removed '\$', commas and whitespace, converted to float, removed outliers. Used in raw and in logarithmic form	240.89	181.50	1,558.00	26.0
		Min. Nights	Minimum night requirement can create bins for nonlinear effects	15.50	23.35	270.0	1.0
	Property Attributes	Accommodates	Number of people the listing can accommodate	3.52	2.54	16.0	1.0
		Bedrooms	Number of bedrooms, used to complement 'Accommodates'	1.47	1.14	8.0	0
		Beds	Number of beds, used to complement 'Accommodates'	1.97	1.59	20.0	0
Control	Host Attributes	Superhost Status	Whether a host is a 'Superhost' or not on the Airbnb platform. Converted to a binary (1=superhost, 0=non-superhost)	0.46	0.50	1.0	0
		Response Rate	Response rate of the host to customer messages, transformed from percentage to a float	0.97	0.11	1.0	0
		Acceptance Rate	Rate of accepted bookings by the listing, transformed from percentage to a float	0.92	0.19	1.0	0
	Review Performance	Review Count	Number of reviews on the listing, can signal popularity, used in raw and in logarithmic form	78.64	114.59	989.0	1.0
		Review Quality	Average rating of all reviews on the listing, shows perceived quality	4.73	0.41	5.0	1.0
	Listing Attributes	Instant Booking	Whether the listing allows for instant booking or not, converted to a binary (1=yes, 0=no)	0.39	0.49	1.0	0
	Location Attributes	Neighborhood	Which neighborhood the listing is located, can signal desirable neighborhoods	-	-	-	-

Statistical Modeling

Variable Group	Independent Variable (IV)	DV: Revenue L365D		
		Model (1) (With only control variables)	Model (2) (Add main variable and moderators)	Model (3) (Add all moderation / interaction effects)
Control Variables	Constant/Intercept	-2.451 (1.331)	-0.983 (1.434)	-0.456 (1.590)
	Superhost Status	0.806 (0.140) ***	0.628 (0.135) ***	0.623 (0.139) ***
	Response Rate	4.949 (0.815) ***	4.614 (0.831) ***	4.747 (0.834) ***
	Instant Booking	0.512 (0.134) ***	0.048 (0.131)	0.048 (0.131)
	Review Count (log)	0.951 (0.047) ***	0.718 (0.050) ***	0.715 (0.050) ***
	Review Quality	0.508 (0.224) *	0.487 (0.213) *	0.459 (0.214) *
	Neighborhood - Dorchester	-0.099 (0.245)	-0.299 (0.255)	-0.310 (0.255)
	Neighborhood - Downtown	0.693 (0.297) *	0.511 (0.295)	3.372 (3.724)
	Neighborhood - Roxbury	-0.881 (0.306) **	-0.834 (0.289) **	2.712 (2.948)
	Neighborhood - Other	-0.083 (0.193)	-0.321 (0.194)	-0.288 (0.193)
Main Effects				
Main Variable(s) and Moderators	Price (log)		0.043 (0.144)	-0.054 (0.187)
	Minimum Nights		-0.037 (0.005) ***	-0.029 (0.007) ***
	Accommodates		0.148 (0.033) ***	-0.197 (0.217)
Moderating Effects/Interactions				
Interactions	Price x Accommodates			0.057 (0.037)
	Min. Nights x Accommodates			-0.001 (0.002)
	Price x Downtown			-0.730 (0.698)
	Price x Roxbury			-0.574 (0.665)
	Min. Nights x Downtown			-0.019 (0.029)
	Min. Nights x Roxbury			-0.039 (0.011) ***
	Accommodates x Downtown			0.452 (0.147) **
	Accommodates x Roxbury			0.038 (0.140)
Additional Results				
Observations	N.A.	2,499	2,499	2,499
Adjusted R ²	N.A.	0.259	0.319	0.327

Note: Standard errors in parentheses. *p<0.05, **p<0.01, ***p<0.001

Analysis and Discussion

The regression results reveal several meaningful patterns regarding factors influencing Airbnb revenue in Boston. Across all three models, host-related controls such as being a superhost, having a high response rate, and receiving many reviews exhibit significant positive effects on revenue, consistent with the idea from Signaling Theory that guests interpret these host characteristics as signals of reliability and quality. Neighborhood effects are also pronounced: Downtown listings consistently show higher revenue, while Roxbury listings have negative coefficients in the baseline models, indicating location-specific demand differences. These findings underscore that location and host reputation are key determinants of Airbnb performance. Regarding main independent variables, minimum-night requirements and accommodation size produce nuanced effects. In Model 2, 'Minimum Nights' is significantly negative, suggesting that longer stay requirements reduce revenue, likely by limiting booking flexibility. 'Accommodates' is positive in Model 2 but becomes negative and nonsignificant when interactions are introduced in Model 3, indicating that the revenue benefit of larger listings depends on other factors. Price, surprisingly, is not statistically significant across models, which may reflect competitive market dynamics where higher prices do not automatically translate into higher revenue after controlling for host quality, location, and review signals.

The interactions in Model 3 provide additional insight. While some interactions with neighborhoods (such as Price_x_Downtown) are not significant, the interaction between minimum nights and Roxbury is significantly negative, suggesting that imposing long minimum-night requirements in lower-demand neighborhoods exacerbates revenue losses. Notably, the interaction between accommodates and Downtown is significantly positive, implying that larger listings in high-demand areas can capture additional revenue, highlighting the importance of considering both listing size and neighborhood demand jointly.

Pricing strategies and minimum-night requirements affect revenue differently depending on listing characteristics and location, and listing size has a conditional effect moderated by neighborhood demand. Signaling Theory helps explain why host quality and visibility metrics remain highly significant even after controlling for price and size. Guests use these cues to assess the reliability and value of listings in an opaque market.

From a managerial perspective, these findings suggest actionable strategies. Hosts should prioritize maintaining high response rates, achieving superhost status, and encouraging guest reviews to signal quality. Listing properties in high-demand neighborhoods like Downtown can enhance revenue, and careful consideration of minimum-night requirements can prevent lost bookings. For larger listings, targeting high-demand neighborhoods can yield higher returns, while smaller properties may perform more consistently in moderate-demand areas. Real-world decision-makers can use these insights to optimize listing characteristics, pricing, and booking policies to improve occupancy and revenue outcomes, tailoring strategies to both property features and neighborhood-specific demand.